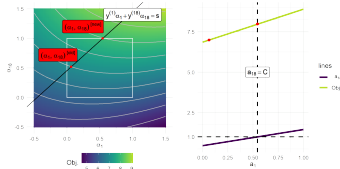
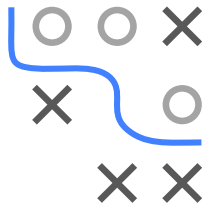


Introduction to Machine Learning

Linear Support Vector Machines

Support Vector Machine Training

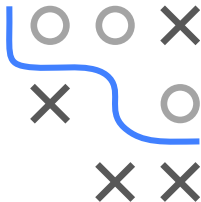


Learning goals

- Know that the SVM problem is not differentiable
- Know how to optimize the SVM problem in the primal via subgradient descent
- Know how to optimize SVM in the dual formulation via pairwise coordinate ascent

SUPPORT VECTOR MACHINE TRAINING I

- Until now, we have ignored the issue of solving the various convex optimization problems.
- The first question is whether we should solve the **primal** or the **dual problem**.
- In the literature SVMs are usually trained in the dual.
- However, SVMs can be trained both in the primal and the dual – each approach has its advantages and disadvantages.
- It is not easy to create an efficient SVM solver, and often specialized approaches have been developed, we only cover basic ideas here.



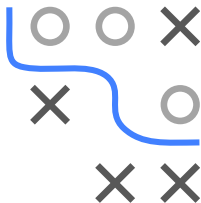
PEGASOS: SSGD IN THE PRIMAL I

Approximate the risk by a stochastic 1-sample version:

$$\frac{\lambda}{2} \|\boldsymbol{\theta}\|^2 + L\left(y^{(i)}, f(\mathbf{x}^{(i)} | \boldsymbol{\theta})\right)$$

With: $f(\mathbf{x} | \boldsymbol{\theta}) = \boldsymbol{\theta}^T \mathbf{x} + \theta_0$ and $L(y, f(\mathbf{x})) = \max(0, 1 - yf)$

The subgradient for $\boldsymbol{\theta}$ is $\lambda \boldsymbol{\theta} - y^{(i)} \mathbf{x}^{(i)} \mathbb{I}_{yf < 1}$



Stochastic subgradient descent (without intercept θ_0)

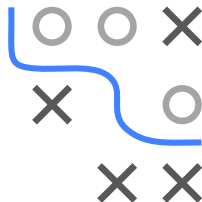
- 1: **for** $t = 1, 2, \dots$ **do**
 - 2: Pick step size α
 - 3: Randomly pick an index i
 - 4: If $y^{(i)} f(\mathbf{x}^{(i)}) < 1$ set $\boldsymbol{\theta}^{[t+1]} = (1 - \lambda\alpha)\boldsymbol{\theta}^{[t]} + \alpha y^{(i)} \mathbf{x}^{(i)}$
 - 5: If $y^{(i)} f(\mathbf{x}^{(i)}) \geq 1$ set $\boldsymbol{\theta}^{[t+1]} = (1 - \lambda\alpha)\boldsymbol{\theta}^{[t]}$
 - 6: **end for**
-

Note the weight decay due to the L2-regularization.

TRAINING SVM IN THE DUAL I

The dual problem of the soft-margin SVM is

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y^{(i)} y^{(j)} \langle \mathbf{x}^{(i)}, \mathbf{x}^{(j)} \rangle \\ \text{s.t.} \quad & 0 \leq \alpha_i \leq C \quad \sum_{i=1}^n \alpha_i y^{(i)} = 0 \end{aligned}$$



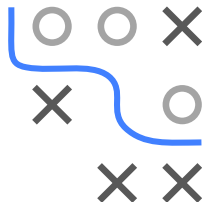
We could solve this problem using coordinate ascent. That means we optimize w.r.t. α_1 , for example, while holding $\alpha_2, \dots, \alpha_n$ fixed.

But: We cannot make any progress since α_1 is determined by $\sum_{i=1}^n \alpha_i y^{(i)} = 0$!

TRAINING SVM IN THE DUAL II

Solution: Update two variables simultaneously

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y^{(i)} y^{(j)} \langle \mathbf{x}^{(i)}, \mathbf{x}^{(j)} \rangle \\ \text{s.t.} \quad & 0 \leq \alpha_i \leq C \quad \sum_{i=1}^n \alpha_i y^{(i)} = 0 \end{aligned}$$



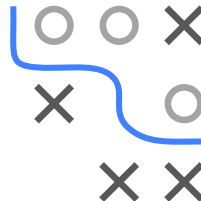
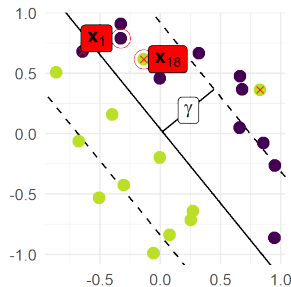
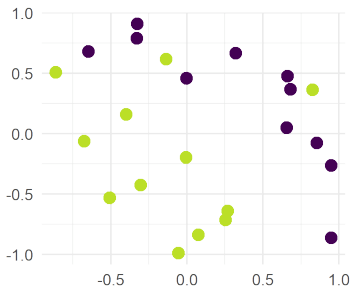
Pairwise coordinate ascent in the dual

- 1: Initialize $\alpha = 0$ (or more cleverly)
 - 2: **for** $t = 1, 2, \dots$ **do**
 - 3: Select some pair α_i, α_j to update next
 - 4: Optimize dual w.r.t. α_i, α_j , while holding α_k ($k \neq i, j$) fixed
 - 5: **end for**
-

The objective is quadratic in the pair, and $s := y^{(i)}\alpha_i + y^{(j)}\alpha_j$ must stay constant. So both α are changed by same (absolute) amount, the signs of the change depend on the labels.

TRAINING SVM IN THE DUAL III

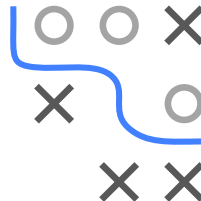
Assume we are in a valid state, $0 \leq \alpha_j \leq C$. Then we chose¹ two observations (encircled in red) for the next iteration. Note they have opposite labels so the sign of their change is equal.



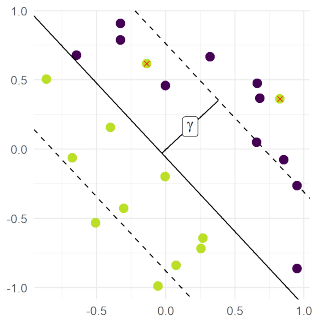
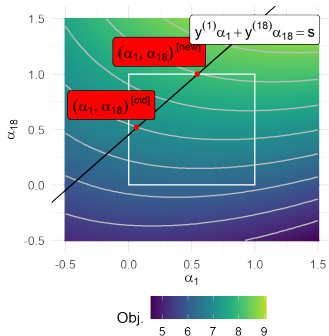
¹There are heuristics to pick the observations to speed up convergence.

TRAINING SVM IN THE DUAL I

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y^{(i)} y^{(j)} \langle \mathbf{x}^{(i)}, \mathbf{x}^{(j)} \rangle \\ \text{s.t.} \quad & 0 \leq \alpha_i \leq C \quad \sum_{i=1}^n \alpha_i y^{(i)} = 0 \end{aligned}$$



We move on the linear constraint until the pair-optimum or the boundary (here: $C = 1$).



TRAINING SVM IN THE DUAL II

Sequential Minimal Optimization (SMO) exploits the fact that effectively we only need to solve a one-dimensional quadratic problem, over in interval, for which an analytical solution exists.

